

9 Answer: C

Work done to stretch it 5.0 cm: $\frac{1}{2} k (0.05)^2 = 5$ $K = 4000 \text{ N/m}$

Work done to stretch by further 5 cm, i.e. $x = 10 \text{ cm}$:

$$\frac{1}{2} (4000) (0.10)^2 = 20 \text{ J}$$

$$\text{Additional work required} = 20 - 5 = 15 \text{ J}$$